

Assignment No. 1

Natural Language Processing BS Software Engineering - 5th Semester - Section B

Deadline: Friday, 8 November

Objective: Use the Naive Bayes classifier to make predictions based on the provided datasets.

Dataset 1: Animals Information

Index	Animals	Size of Animal	Body Color	Can we Pet them
0	Dog	Medium	Black	Yes
1	Dog	Big	White	No
2	Rat	Small	White	Yes
3	Cow	Big	White	Yes
4	Cow	Small	Brown	No
5	Cow	Big	Black	Yes
6	Rat	Big	Brown	No
7	Dog	Small	Brown	Yes
8	Dog	Medium	Brown	Yes
9	Cow	Medium	White	No
10	Dog	Small	Black	Yes
11	Rat	Medium	Black	No
12	Rat	Small	Brown	No
13	Cow	Big	White	Yes

Assignment Condition:

- Predict whether a Medium sized animal that is Brown in color can be petted.

Dataset 2: Weather and Road Conditions

S.No	Weather Condition	Road Condition	Traffic Condition	Engine Problem	Accident
1	Rain	Bad	High	No	Yes
2	Snow	Average	Normal	Yes	No
3	Clear	Bad	Light	No	No
4	Clear	Good	Light	Yes	Yes
5	Snow	Good	Normal	No	No
6	Rain	Average	Light	No	No
7	Rain	Good	Normal	No	No
8	Snow	Bad	High	No	Yes
9	Clear	Good	High	Yes	No
10	Clear	Bad	High	Yes	Yes

Assignment Condition:

- Based on the following conditions, predict whether an accident will occur:
 - Weather Condition: Rain, Road Condition: Good, Traffic Condition: Light, Engine Problem: No

Dataset 3: Loan Approval

ID	Age Group	Income Level	Credit Score	Employment Status	Loan Approved
1	Young	High	Good	Employed	Yes
2	Middle	Low	Poor	Unemployed	No
3	Senior	High	Excellent	Retired	Yes
4	Young	Low	Good	Employed	No
5	Middle	High	Poor	Employed	Yes
6	Senior	Low	Excellent	Retired	No
7	Young	High	Poor	Employed	No
8	Middle	High	Excellent	Employed	Yes
9	Senior	Low	Good	Retired	No
10	Young	Low	Excellent	Employed	Yes

Assignment Condition:

- Predict whether a loan will be approved for a Middle-aged person with High income, Good credit score, and Employed status.

Dataset 4: Email Spam Detection

Email ID	Contains Offer	Contains Link	Contains Greeting	Sender Known	Spam
101	Yes	Yes	Yes	No	Yes
102	No	No	Yes	Yes	No
103	Yes	Yes	No	No	Yes
104	No	Yes	Yes	Yes	No
105	Yes	No	No	No	Yes
106	No	No	Yes	No	No
107	Yes	Yes	Yes	Yes	No
108	No	Yes	No	No	Yes
109	Yes	No	Yes	Yes	No
110	No	No	No	No	Yes

Assignment Condition:

- Predict whether an email with the following attributes will be classified as spam:
 - Contains Offer: Yes, Contains Link: No, Contains Greeting: Yes, Sender Known: No

Submission Guidelines:

- Perform calculations using the Naive Bayes formula.
- Present your findings clearly, indicating the predicted outcomes for each condition.
- Discuss any patterns or insights you observe from the data.